



Advancing Water Quality Assessment and Monitoring with a Robust Stacked Ensemble Method

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Abstract

Water quality monitoring is crucial in assessing the health of surface water bodies and aquifers, ensuring water safety for various purposes including drinking, agriculture, and ecosystem support. Traditional water quality monitoring relies on established methods and protocols. As a common practice, the Water Quality Index (WQI) is used to summarize and communicate the overall quality of water based on multiple water quality parameters. Some studies employ statistical analysis to identify trends, anomalies, and correlations in water quality data. However, practical adoption of machine learning in water quality monitoring systems remains rare. This study integrates a machine learning algorithm with the WQI to create a predictive model. We have proposed an ensemble model that significantly outperforms all individual algorithms, achieving the lowest Mean Square Error (MSE), Mean Absolute Error (MAE) and a perfect r-squared value of 1, indicating its superior ability to predict water quality. This novel stacked ensemble machine learning algorithm enables real-time or near-real-time assessments of water quality by leveraging specific water quality parameters. This model was tested in selected lakes in southern India and demonstrate its capability to forecast and analyze water quality parameters across various aquatic environments globally.

Keywords Water quality index · Machine learning · Regression analysis · Stacked ensemble techniques · Regression accuracy

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